CHAPTER 19

the Heath, Jarrow & Morton and Brace, Gatarek & Musiela models

The aim of this Chapter...

...is to explain in as simple a way as possible the breakthroughs in interest rate modeling known as the Heath, Jarrow & Morton model and the Brace, Gatarek & Musiela model. Unfortunately, even the simplest explanation is tough going and I would understand if you skipped all the math in this chapter.

In this Chapter...

- the Heath, Jarrow & Morton forward rate model
- the relationship between HJM and spot rate models
- the advantages and disadvantages of the HJM approach
- how to decompose the random movements of the forward rate curve into its principal components
- the Brace, Gatarek & Musiela model

19.1 INTRODUCTION

The **Heath, Jarrow & Morton** approach to the modeling of the whole forward rate curve was a major breakthrough in the pricing of fixed-income products. They built up a framework that encompassed all of the models we have seen so far (and many that we haven't). Instead of modeling a short-term interest rate and deriving the forward rates (or, equivalently, the yield curve) from that model, they boldly start with a model for the whole forward rate curve. Since the forward rates are known today, the matter of yield curve fitting is contained naturally within their model, it does not appear as an afterthought. Moreover, it is possible to take *real data* for the random movement of the forward rates and incorporate them into the derivative pricing methodology.

19.2 THE FORWARD RATE EQUATION

The key concept in the HJM model is that we model the evolution of the whole forward rate curve, not just the short end. Write $F(t;T)$ for the forward rate curve at time t. Thus the price of a zero-coupon bond at time t and maturing at time T, when it pays \$1, is

$$Z(t;T) = e^{-\int_t^T F(t;s)\,ds}. \tag{19.1}$$

Let us assume that all zero-coupon bonds evolve according to

$$dZ(t;T) = \mu(t,T)Z(t;T)dt + \sigma(t,T)Z(t;T)dX. \tag{19.2}$$

This is not much of an assumption, other than to say that it is a one-factor model, and I will generalize that later. In this $d\cdot$ means that time t evolves but the maturity date T is fixed. Note that since $Z(t;t) = 1$ we must have $\sigma(t,t) = 0$. From (19.1) we have

$$F(t;T) = -\frac{\partial}{\partial T}\log Z(t;T).$$

Differentiating this with respect to t and substituting from (19.2) results in an equation for the evolution of the forward curve:

$$dF(t;T) = \frac{\partial}{\partial T}\left(\tfrac{1}{2}\sigma^2(t,T) - \mu(t,T)\right)dt - \frac{\partial}{\partial T}\sigma(t,T)dX. \tag{19.3}$$

In Figure 19.1 is shown the forward rate curve today, time t^*, and a few days later. The whole curve has moved according to (19.3).

Where has this got us? We have an expression for the drift of the forward rates in terms of the volatility of the forward rates. There is also a μ term, the drift of the bond. We have seen many times before how such drift terms disappear when we come to pricing derivatives, to be replaced by the risk-free interest rate r. Exactly the same will happen here.

19.3 THE SPOT RATE PROCESS

The spot interest rate is simply given by the forward rate for a maturity equal to the current date, i.e.

$$r(t) = F(t;t).$$

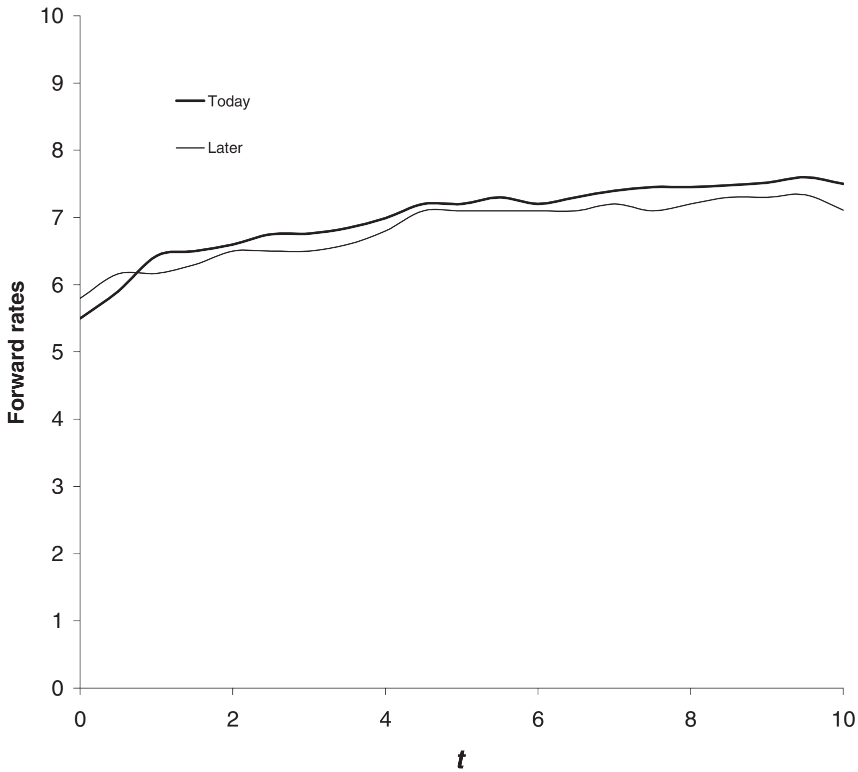


Figure 19.1 The forward rate curve today and a few days later.

In this section I am going to manipulate this expression to derive the stochastic differential equation for the spot rate. In so doing we will begin to see why the HJM approach can be slow to price derivatives.

Suppose today is t^* and that we know the whole forward rate curve today, $F(t^*; T)$. We can write the spot rate for *any* time t in the future as

$$r(t) = F(t; t) = F(t^*; t) + \int_{t^*}^{t} dF(s; t).$$

From our earlier expression (19.3) for the forward rate process for F we have

$$r(t) = F(t^*; t) + \int_{t^*}^{t} \left(\sigma(s, t) \frac{\partial \sigma(s, t)}{\partial t} - \frac{\partial \mu(s, t)}{\partial t} \right) ds - \int_{t^*}^{t} \frac{\partial \sigma(s, t)}{\partial t} dX(s).$$

Differentiating this with respect to time t we arrive at the stochastic differential equation for r

$$dr = \left(\frac{\partial F(t^*; t)}{\partial t} - \frac{\partial \mu(t, s)}{\partial s} \Big|_{s=t} + \int_{t^*}^{t} \left(\sigma(s, t) \frac{\partial^2 \sigma(s, t)}{\partial t^2} + \left(\frac{\partial \sigma(s, t)}{\partial t} \right)^2 - \frac{\partial^2 \mu(s, t)}{\partial t^2} \right) ds \right.$$
$$\left. - \underline{\int_{t^*}^{t} \frac{\partial^2 \sigma(s, t)}{\partial t^2} dX(s)} \right) dt - \frac{\partial \sigma(t, s)}{\partial s} \Big|_{s=t} dX.$$

Time Out...

Eeeek!

This is not a pretty sight. The idea is simple but the math is not. Hold the thought that in the HJM model we move/model *the whole of the forward rate curve*, and not just one end of it. Since we start with today's forward rate curve we don't have to worry about getting discount factors correct initially, they are automatically correct.

If you find the math daunting, just read the words for the rest of this chapter.

19.3.1 The non-Markov nature of HJM

The details of this expression are not important. I just want you to observe one point. Compare this stochastic differential equation for the spot rate with any of the models in Chapter 16. Clearly, it is more complicated, there are many more terms. All but the last one are deterministic, the last is random. The important point concerns the nature of these terms. In particular, the term underlined depends on the history of σ from the date t^* to the future date t, and *it depends on the history of the stochastic increments dX*. This term is thus highly path dependent. Moreover, for a general HJM model it makes the motion of the spot rate **non-Markov**. In a **Markov process** it is only the present state of a variable that determines the possible future (albeit random) state. Having a non-Markov model may not matter to us if we can find a small number of extra state variables that contain all the information that we need for predicting the future. Unfortunately, the general HJM model requires an infinite number of such variables to define the present state; if we were to write the HJM model as a partial differential equation we would need an infinite number of independent variables.

At the moment we are in the real world. To price derivatives we need to move over to the risk-neutral world. The first step in this direction is to see what happens when we hold a hedged portfolio.

19.4 THE MARKET PRICE OF RISK

In the one-factor HJM model all stochastic movements of the forward rate curve are perfectly correlated. We can therefore hedge one bond with another of a different maturity. Such a hedged portfolio is

$$\Pi = Z(t;T_1) - \Delta Z(t;T_2).$$

The change in this portfolio is given by

$$\begin{aligned} d\Pi &= dZ(t;T_1) - \Delta\, dZ(t;T_2) \\ &= Z(t;T_1)\left(\mu(t,T_1)dt + \sigma(t,T_1)dX\right) - \Delta Z(t;T_2)\left(\mu(t,T_2)dt + \sigma(t,T_2)dX\right). \end{aligned}$$

If we choose

$$\Delta = \frac{\sigma(t, T_1)Z(t; T_1)}{\sigma(t, T_2)Z(t; T_2)}$$

then our portfolio is hedged, is risk free. Setting its return equal to the risk-free rate $r(t)$ and rearranging we find that

$$\frac{\mu(t, T_1) - r(t)}{\sigma(t, T_1)} = \frac{\mu(t, T_2) - r(t)}{\sigma(t, T_2)}.$$

The left-hand side is a function of T_1 and the right-hand side is a function of T_2. This is only possible if both sides are independent of the maturity date T:

$$\mu(t, T) = r(t) + \lambda(t)\sigma(t, T).$$

As before, $\lambda(t)$ is the market price of risk (associated with the one factor).

19.5 REAL AND RISK NEUTRAL

We are almost ready to price derivatives using the HJM model. But first we must discuss the real and risk-neutral worlds, relating them to the ideas in previous chapters.

All of the variables I have introduced above have been *real* variables. But when we come to pricing derivatives we must do so in the risk-neutral world. In the present HJM context, risk-neutral 'means' $\mu(t, T) = r(t)$. This means that in the risk-neutral world the return on any traded investment is simply $r(t)$. We can see this in (19.2). The risk-neutral zero-coupon bond price satisfies

$$dZ(t; T) = r(t)Z(t; T)dt + \sigma(t, T)Z(t; T)dX.$$

The deterministic part of this equation represents exponential growth of the bond at the risk-free rate. The form of the equation is very similar to that for a risk-neutral equity, except that here the volatility will be much more complicated.

19.5.1 The relationship between the risk-neutral forward rate drift and volatility

Let me write the stochastic differential equation for the *risk-neutral* forward rate curve as

$$dF(t; T) = m(t, T)dt + \nu(t, T)dX.$$

From (19.3)

$$\nu(t, T) = -\frac{\partial}{\partial T}\sigma(t, T)$$

is the forward rate volatility and, from (19.3), the drift of the forward rate is given by

$$\frac{\partial}{\partial T}\left(\tfrac{1}{2}\sigma^2(t, T) - \mu(t, T)\right) = \nu(t, T)\int_t^T \nu(t, s)ds - \frac{\partial}{\partial T}\mu(t, T),$$

where we have used $\sigma(t,t) = 0$. In the risk-neutral world we have $\mu(t,T) = r(t)$, and so the drift of the risk-neutral forward rate curve is related to its volatility by

$$m(t,T) = \nu(t,T)\int_t^T \nu(t,s)ds. \tag{19.4}$$

And so

$$dF(t;T) = \nu(t,T)\left(\int_t^T \nu(t,s)ds\right) dt + \nu(t,T)dX. \tag{19.5}$$

19.6 PRICING DERIVATIVES

Pricing derivatives is all about finding the expected present value of all cashflows in a risk-neutral framework. This was discussed in Chapter 6, in terms of equity, currency and commodity derivatives. If we are lucky then this calculation can be done via a low-dimensional partial differential equation. The HJM model, however, is a very general interest rate model and in its full generality one cannot write down a finite-dimensional partial differential equation for the price of a derivative.

Because of the non-Markov nature of HJM in general a partial differential equation approach is infeasible. This leaves us with two alternatives. One is to estimate directly the necessary expectations by simulating the random evolution of, in this case, the risk-neutral forward rates. The other is to build up a tree structure.

19.7 SIMULATIONS

If we want to use a Monte Carlo method, then we must simulate the evolution of the whole forward rate curve, calculate the value of all cashflows under each evolution and then calculate the present value of these cashflows by *discounting at the realized spot rate, $r(t)$*.

To price a derivative using a Monte Carlo simulation, perform the following steps. I will assume that we have chosen a model for the forward rate volatility, $\nu(t,T)$. Today is t^* when we know the forward rate curve $F(t^*;T)$.

1. Simulate a realized evolution of the whole risk-neutral forward rate curve for the necessary length of time, until T^*, say. This requires a simulation of

   $$dF(t;T) = m(t,T)dt + \nu(t,T)dX,$$

   where

   $$m(t,T) = \nu(t,T)\int_t^T \nu(t,s)ds.$$

   After this simulation we will have a realization of $F(t;T)$ for $t^* \le t \le T^*$ and $T \ge t$. We will have a realization of the whole forward rate path.

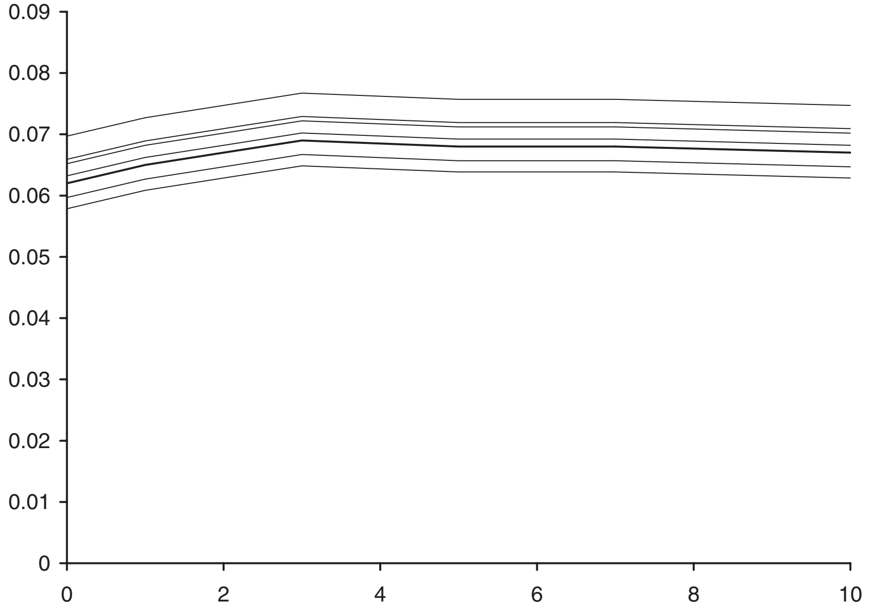
2. At the end of the simulation we will have the realized prices of all maturity zero-coupon bonds at every time up to T^*.

3. Using this forward rate path calculate the value of all the cashflows that would have occurred.
4. Using the realized path for the spot interest rate $r(t)$ calculate the present value of these cashflows. Note that we discount at the continuously compounded risk-free rate, not at any other rate. In the risk-neutral world all assets have an expected return of $r(t)$.
5. Return to Step 1 to perform another realization, and continue until you have a sufficiently large number of realizations to calculate the expected present value as accurately as required.

Time Out...

Evolution of the forward curve

In this figure we see how the forward curve may have evolved during one set of simulations.



The disadvantage of the HJM model is that a Monte Carlo simulation such as this can be slow. On the plus side, because the whole forward rate curve is calculated the bond prices at all maturities are trivial to find during this simulation.

19.8 TREES

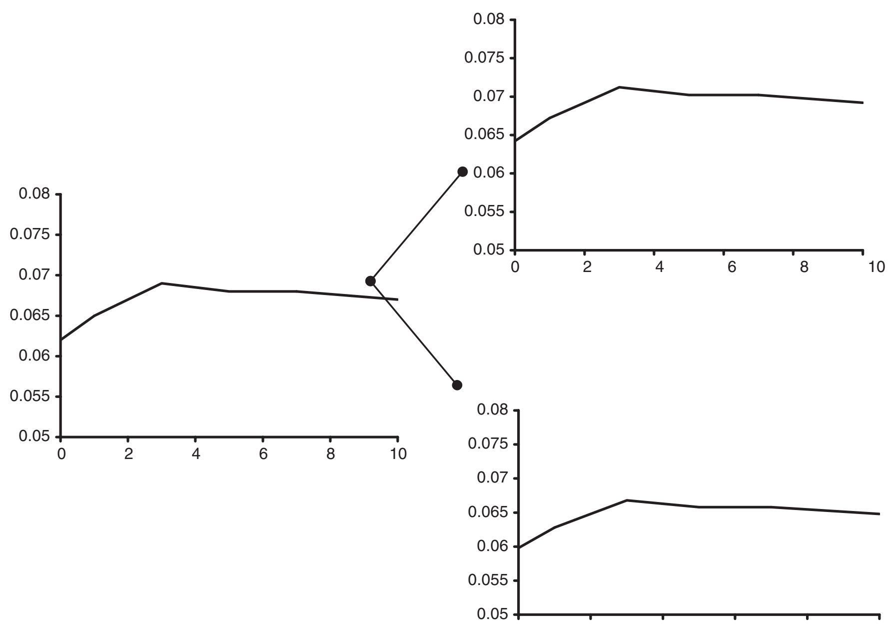
If we are to build up a tree for a non-Markov model then we find ourselves with the unfortunate result that the forward curve after an up move followed by a down is *not* the same as the curve after a down followed by an up. The equivalence of these two paths in the Markov world is what makes the binomial method so powerful and efficient. In the non-Markov world our tree structure becomes 'bushy,' and grows *exponentially* in size with the addition of new time steps.

If the contract we are valuing is European with no early exercise then we don't need to use a tree, a Monte Carlo simulation can be immediately implemented. However, if the contract has some American feature then to correctly price in the early exercise we don't have much choice but to use a tree structure. The exponentially large tree structure will make the pricing problem very slow.

Time Out...

Trees for the whole forward rate curve

The figures below show how the forward rate curve might evolve in a tree-like version of the HJM model.



19.9 THE MUSIELA PARAMETERIZATION

Often in practice the model for the volatility structure of the forward rate curve will be of the form

$$\nu(t,T) = \overline{\nu}(t,T-t),$$

meaning that we will model the volatility of the forward rate at each maturity, one, two, three years, and not at each maturity date, 2006, 2007, ... If we write τ for the maturity period $T-t$ then it is a simple matter to find that $\overline{F}(t;\tau) = F(t,t+\tau)$ satisfies

$$d\overline{F}(t;\tau) = \overline{m}(t,\tau)dt + \overline{\nu}(t,\tau)dX,$$

where

$$\overline{m}(t,\tau) = \overline{\nu}(t,\tau)\int_0^\tau \overline{\nu}(t,s)ds + \frac{\partial}{\partial\tau}\overline{F}(t,\tau).$$

It is much easier in practice to use this representation for the evolution of the risk-neutral forward rate curve.

19.10 MULTI-FACTOR HJM

Often a single-factor model does not capture the subtleties of the yield curve that are important for particular contracts. The obvious example is the spread option that pays off the difference between rates at two different maturities. We then require a multi-factor model. The multi-factor theory is identical to the one-factor case, so we can simply write down the extension to many factors.

If the risk-neutral forward rate curve satisfies the N-dimensional stochastic differential equation

$$dF(t;T) = m(t;T)dt + \sum_{i=1}^{N} \nu_i(t;T)dX_i,$$

where the dX_i are uncorrelated, then

$$m(t,T) = \sum_{i=1}^{N} \nu_i(t,T)\int_t^T \nu_i(t,s)ds.$$

And so

$$dF(t,T) = \left(\sum_{i=1}^{N} \nu_i(t,T)\int_t^T \nu_i(t,s)ds\right) dt + \sum_{i=1}^{N} \nu_i(t,T)dX_i. \quad (19.6)$$

19.11 SPREADSHEET IMPLEMENTATION

The HJM model is very easy to implement on a spreadsheet. Figure 19.2 shows the results of such a simulation for a two-factor model. In this example the Musiela parameterization has been used and $\overline{\nu}$ is a function of $\tau = T - t$ only. This means that the function $\overline{m}$ contains the first volatility term which is just a function of time to maturity (this is row 3 in the figure) and the slope of the forward curve term (this latter is calculated in each of the cells from row 11 down). In this example there are two volatility factors, the first is constant and the second is linear in τ.

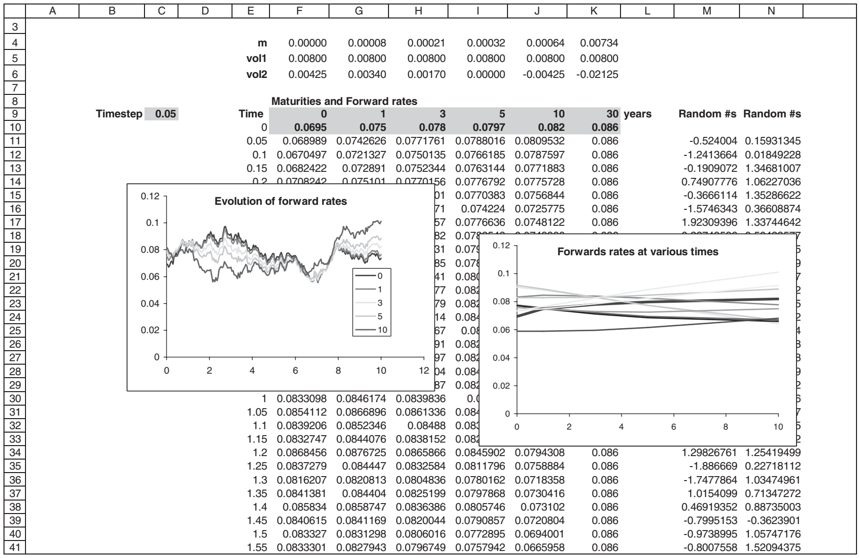


Figure 19.2 Spreadsheet showing results of a two-factor HJM simulation.

19.12 A SIMPLE ONE-FACTOR EXAMPLE: HO & LEE

In this section we make a comparison between the spot rate modeling of Chapter 16 and HJM. One of the key points about the HJM approach is that the yield curve is fitted by default. The simplest yield curve fitting spot rate model is Ho & Lee, so we draw a comparison between this and HJM.

In Ho & Lee the risk-neutral spot rate satisfies

$$dr = \eta(t)dt + c\,dX,$$

for a constant c. The prices of zero-coupon bonds, $Z(r, t; T)$, in this model satisfy

$$\frac{\partial Z}{\partial t} + \tfrac{1}{2}c^2\frac{\partial^2 Z}{\partial r^2} + \eta(t)\frac{\partial Z}{\partial r} - rZ = 0$$

with

$$Z(r, T; T) = 1.$$

The solution is easily found to be

$$Z(r, t; T) = \exp\left(\tfrac{1}{6}c^2(T-t)^3 - \int_t^T \eta(s)(T-s)ds - (T-t)r\right).$$

In the Ho & Lee model $\eta(t)$ is chosen to fit the yield curve at time t^*. In forward rate terms this means that

$$F(t^*; T) = r(t^*) - \tfrac{1}{2}c^2(T-t^*)^2 + \int_{t^*}^T \eta(s)ds,$$

and so

$$\eta(t) = \frac{\partial F(t^*; t)}{\partial t} + c^2(t - t^*).$$

At any time later than t^*

$$F(t; T) = r(t) - \tfrac{1}{2}c^2(T - t)^2 + \int_t^T \eta(s)ds.$$

From this we find that

$$dF(t; T) = c^2(T - t)dt + c\,dX.$$

In our earlier notation, $\sigma(t, T) = -c(T - t)$ and $\nu(t, T) = c$. This is the evolution equation for the risk-neutral forward rates. It is easily confirmed for this model that Equation (19.4) holds. This is the HJM representation of the Ho & Lee model. Most of the popular models have HJM representations.

19.13 PRINCIPAL COMPONENT ANALYSIS

There are two main ways to use HJM. One is to choose the volatility structure $\nu_i(t, T)$ to be sufficiently 'nice' to make a tractable model, one that is Markov. This usually leads us back to the 'classical' popular spot-rate models. The other way is to choose the volatility structure to match data. This is where principal component analysis (PCA) comes in.

In analyzing the volatility of the forward rate curve one usually assumes that the volatility structure depends only on the time to maturity, i.e.

$$\nu = \overline{\nu}(T - t).$$

I will assume this but examine a more general multi-factor model:

$$dF(t; T) = m(t, T)dt + \sum_{i=1}^{N} \overline{\nu}_i(T - t)dX_i.$$

From time series data we can determine the functions $\overline{\nu}_i$ empirically, this is **principal components analysis**. I will give a loose description of how this is done, with more details in the spreadsheets.

If we have forward rate time series data going back a few years we can calculate the covariances between the *changes* in the rates of different maturities. We may have, for example, the one-, three-, six-month, one-, two, three-, five-, seven-, 10- and 30-year rates. The covariance matrix would then be a 10×10 symmetric matrix with the variances of the rates along the diagonal and the covariances between rates off the diagonal.

In Figure 19.3 is shown a spreadsheet of daily one-, three- and six-month rates, and the day-to-day changes. The covariance matrix for these changes is also shown.

PCA is a technique for finding common movements in the rates, for essentially finding eigenvalues and eigenvectors of the matrix. We expect to find, for example, that a large part of the movement of the forward rate curve is common between rates, that a parallel

	A	B	C	D	E	F	G	H	I	J	K
1	**Forward rates:**				**Changes in rates:**						
2		**1 month**	**3 month**	**6 month**	**1 month**	**3 month**	**6 month**				
3	22-Sep-88	8.25000	8.31250	8.56250							
4	23-Sep-88	8.25000	8.31250	8.56250	0.00000	0.00000	0.00000				
5	26-Sep-88	8.31250	8.37500	8.62500	0.06250	0.06250	0.06250				
6	27-Sep-88	8.31250	8.43750	8.6	0000	0.06250	0.06250		**1 month**	**3 month**	**6 month**
7	28-Sep-88	8.42188	8.50000	8.81250	0.10938	0.06250	0.12500	**1 month**	0.007501		
8	29-Sep-88	8.37500	8.68750	8.81250	-0.04688	0.18750	0.00000	**3 month**	0.003831	0.004225	
9	30-Sep-88	8.31250	8.62500	8.75000	-0.06250	-0.06250	-0.06250	**6 month**	0.003628	0.004020	0.004997
10	3-Oct-88	8.31250	8.62500	8.68750	0.00000	0.00000	-0.06250				
11	4-Oct-88	8.31250	8.56250	8.68750	0.00000	-0.06250	0.00000	**Scaled covariance matrix:**			
12	5-Oct-88	8.31250	8.56250	8.68750	0.00000	0.00000	0.00000		**1 month**	**3 month**	**6 month**
13	6-Oct-88	8.31250	8.56250	8.68750	0.00000	0.00000	0.00000	**1 month**	0.000189		
14	7-Oct-88	8.31250	8.62500	8.75000	0.00000	0.06250	0.06250	**3 month**	0.000097	0.000106	
15	10-Oct-88	8.25000	8.56250	8.56250	-0.06250	-0.06250	-0.18750	**6 month**	0.000091	0.000101	0.000126
16	11-Oct-88	8.25000	8.56250	8.62500	0.00000	0.00000	0.06250				
17	12-Oct-88	8.31250	8.62500	8.68750	0.06250	0.06250	0.06250				
18	13-Oct-88	8.31250	8.64063	8.68750	0.00000	0.01563	0.00000				
19	14-Oct-88	8.31250	8.62500	8.62500	0.00000	-0.01563	-0.06250				
20	17-Oct-88	8.31250	8.62500	8.62500	0.00000	0.00000	0.00000				
21	18-Oct-88	8.31250	8.62500	8.62500	0.00000	0.00000	0.00000				
22	19-Oct-88	8.31250	8.62500	8.62500	0.00000	0.00000	0.00000				
23	20-Oct-88	8.37500	8.68750	8.68750	0.06250	0.06250	0.06250				
24	21-Oct-88	8.37500	8.68750	8.68750	0.00000	0.00000	0.00000				
25	24-Oct-88	8.37500	8.68750	8.75000	0.00000	0.00000	0.06250				
26	25-Oct-88	8.37500	8.68750	8.75000	0.00000	0.00000	0.00000				
27	26-Oct-88	8.37500	8.68750	8.75000	0.00000	0.00000	0.00000				
28	27-Oct-88	8.37500	8.68750	8.68750	0.00000	0.00000	-0.06250				

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Figure 19.3 One-, three- and six-month rates and the changes.

shift in the rates is the largest component of the movement of the curve in general. The next most important movement would be a twisting of the curve, followed by a bending.

Suppose that we have found the covariance matrix, $\mathbf{M}$, for the changes in the rates mentioned above. This 10 by 10 matrix will have 10 eigenvalues, λ_i, and eigenvectors, $\mathbf{v}_i$, satisfying

$$\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i;$$

$\mathbf{v}_i$ is a column vector.

The eigenvector associated with the largest eigenvalue is the first principal component. It gives the dominant part in the movement of the forward rate curve. Its first entry represents the movement of the one-month rate, the second entry is the three-month rate, etc. Its eigenvalue is the variance of these movements. In Figure 19.4 we see the entries in this first principal component plotted against the relevant maturity. This curve is relatively flat when compared with the other components. This indicates that, indeed, a parallel shift of the yield curve is the dominant movement. Note that the eigenvectors are orthogonal, there is no correlation between the principal components.

In this figure are also plotted the next two principal components. Observe that one gives a twisting of the curve and the other a bending.

The result of this analysis is that the volatility factors are given by

$$\overline{\nu}_i(\tau_j) = \sqrt{\lambda_i}(\mathbf{v}_i)_j.$$

Here τ_j is the maturity, i.e. 1/12, 1/4, etc. and $(\mathbf{v}_i)_j$ is the jth entry in the vector $\mathbf{v}_i$. To get the volatility of other maturities will require some interpolation.

The calculation of the covariance matrix is simple, discussed in Chapter 12. The calculation of the eigenvalues and vectors is also simple if you use the following algorithm.

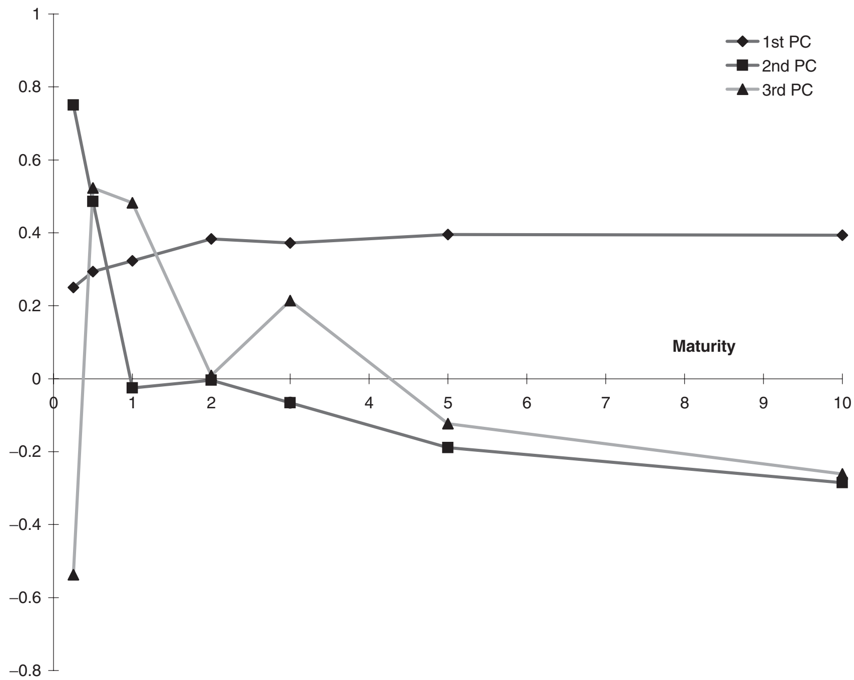


Figure 19.4 The first three principal components for the US forward rate curve. The data run from 1988 until 1996.

19.13.1 The power method

I will assume that all the eigenvalues are distinct, a reasonable assumption given the empirical nature of the matrix. Since the matrix is symmetric positive definite (it is a covariance matrix) we have all the nice properties we need. The eigenvector associated with the *largest* eigenvalue is easily found by the following iterative procedure. First, make an initial guess for the eigenvector, call it $\mathbf{x}^0$. Now iterate using

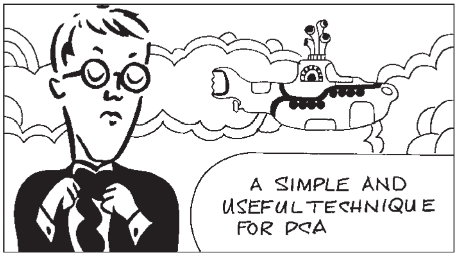
$$\mathbf{y}^{k+1} = \mathbf{M}\mathbf{x}^k,$$

for $k = 0, \ldots,$ and

$$\beta^{k+1} = \text{element of } \mathbf{y}^{k+1} \text{ having largest modulus}$$

followed by

$$\mathbf{x}^{i+1} = \frac{1}{\beta^{k+1}}\mathbf{y}^{k+1}.$$



As $k \to \infty$, $\mathbf{x}^k$ tends to the eigenvector and β^k to the eigenvalue λ. In practice you would stop iterating once you have reached some set tolerance. Thus we have found the first principal component. It is standard to normalize the vector, and this is our $\mathbf{v}_1$.

To find the next principal component we must define a new matrix by

$$\mathbf{N} = \mathbf{M} - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T.$$

Now use the power method on this new matrix **N** to find the second principal component. This process can be repeated until all (10) components have been found.

19.14 OPTIONS ON EQUITIES, ETC.

The pricing of options contingent on equities, currencies, commodities, indices, etc. is straightforward in the HJM framework. All that we need to know are the volatility of the asset and its correlations with the forward rate factors. The Monte Carlo simulation then uses the risk-neutral random walks for both the forward rates and the asset, i.e. zero-coupon bonds and the asset have a drift of $r(t)$. Of course, there are the usual adjustments, to be made for dividends, foreign interest rate or cost of carry, amounting to a change to the drift rate for the asset.

The only fly in the ointment is that American-style exercise is difficult to accommodate.

19.15 NON-INFINITESIMAL SHORT RATE

One of the problems with HJM is that there is no guarantee that interest rates will stay positive, nor that the money market account will stay finite. These problems are associated with the use of a continuously compounded interest rate; all rates can be deduced from the evolution of this rate, but the rate itself is not actually observable. Modeling rates with a finite accruals period, such as three-month LIBOR, for example, have two advantages: the rate is directly observable, and positivity and finiteness can be guaranteed. Let's see how this works. I use the Musiela parameterization of the forward rates.

I have said that $\overline{F}(t, \tau)$ satisfies

$$d\overline{F}(t, \tau) = \ldots + \overline{\nu}(t, \tau) dX.$$

A reasonable choice for the volatility structure might be

$$\overline{\nu}(t, \tau) = \gamma(t, \tau) \overline{F}(t, \tau)$$

for finite, non-zero $\gamma(t, \tau)$. At first sight, this is a good choice for the volatility, after all, lognormality is a popular choice for random walks in finance. Unfortunately, this model leads to exploding interest rates. Yet we would like to retain some form of lognormality of rates, recall that market practice is to assume lognormality of just about everything.

We can get around the explosive rates by defining an interest rate $j(t, \tau)$ that is accrued m times *per annum*. The relationship between the new $j(t, \tau)$ and the old $\overline{F}(t, \tau)$ is then

$$\left(1 + \frac{j(t, \tau)}{m}\right)^m = e^{\overline{F}(t, \tau)}.$$

Now what happens if we choose a lognormal model for the rates? If we choose

$$dj(t, \tau) = \ldots + \gamma(t, \tau) j(t, \tau) dX,$$

it can be shown that this leads to a stochastic differential equation for $\overline{F}(t,\tau)$ of the form

$$d\overline{F}(t,\tau) = \ldots - m\gamma(t,\tau)\left(1 - e^{\overline{F}(t,\tau)/m}\right) dX.$$

The volatility structure in this expression is such that all rates stay positive and no explosion occurs.

If we specify the quantity m, we can of course still do PCA to find out the best form for the function $\gamma(t,\tau) = \gamma(\tau)$.

19.16 THE BRACE, GATAREK & MUSIELA MODEL

The Brace, Gatarek & Musiela[1] model crucially models the actual traded, observable quantities in the fixed-income world. Think of it as a discrete version of the HJM model. This is relatively straightforward, but slightly unnerving if you are used to differential rather than difference equations. The model is popular with practitioners because it can price any contract whose cashflows can be decomposed into functions of the observed forward rates.

We will work in terms of observable, discrete forward rates, i.e. rates that really are quoted in the market, rather than the unrealistic continuous forward curve. With $Z(t;T)$ being the price of a zero-coupon bond at time t that matures at time T we have the forward rates at time t, for the period from T_i to the time T_{i+1}, separated by the time step τ, given by

$$1 + \tau F(t;T_i,T_{i+1}) = 1 + \tau F(t;T_i,T_i+\tau) = 1 + \tau F_i(t) = \frac{Z(t;T_i)}{Z(t;T_{i+1})}.$$

This is the definition of forward rates in terms of zero-coupon bond prices, or vice versa. For simplicity of notation in what follows we shall use F_i to mean $F(t;T_i,T_{i+1})$, and Z_i to mean $Z(t;T_i)$. These are, of course, stochastic quantities, varying with time.

The discount factor in terms of the forward rate is just

$$\frac{1}{1+\tau F_i(t)}.$$

This is the discount factor for present valuing from T_{i+1} back to T_i.

Note that here we are using the discrete compounding definition of an interest rate, consistent with that introduced in Chapter 14, rather than our usual continuous definition that is more often used in the equity world.

Let's suppose that we can write the dynamics of each forward rate, F_i, as

$$dF_i = \mu_i F_i\, dt + \sigma_i F_i\, dX_i.$$

This looks like a lognormal model, but, of course, the μs and σs could be hiding more Fs. So really it's a lot more general than that. Similarly suppose that the zero-coupon bond dynamics are given by

$$dZ_i = rZ_i\, dt + Z_i \sum_{j=1}^{i-1} a_{ij}\, dX_j \tag{19.7}$$

[1] The BGM or BGM/J model, the J standing for Jamshidian.

where

$$Z_i(t) = Z(t; T_i).$$

There are several points to note about this expression. First of all, we are clearly in a risk-neutral world with the drift of the *traded* asset Z being the risk-free rate. If the bond wasn't traded then we couldn't say that its drift was r.[2] That's why we couldn't say that the drift of F_i was r, since the forward rate is *not* a traded instrument.

Actually, we'll see shortly that we don't need a model for r, it will drop out of the analysis. Also, it looks like a lognormal model but again the as could be hiding more Zs.

Finally, the zero-coupon bond volatility is only given in terms of the volatilities of forward rates of shorter maturities. This is because volatility after its maturity will not affect the value of a bond, and that's why there is no a_{ii} term in Equation (19.7).

We can write

$$Z_i = (1 + \tau F_i)Z_{i+1}.$$

Applying Itô's lemma to this we get

$$dZ_i = (1 + \tau F_i)dZ_{i+1} + \tau Z_{i+1}\,dF_i + \tau\sigma_i F_i Z_{i+1}\left(\sum_{j=1}^{i} a_{i+1,j}\rho_{ij}\right) dt, \qquad (19.8)$$

where ρ_{ij} is the correlation between dX_i and dX_j.

Think of Equation (19.8) as containing three types of terms: dX_i; dX_j for $j = 1, \dots, i-1$; dt.

Equating coefficients of dX_i in Equation (19.8) we get

$$0 = (1 + \tau F_i)a_{i+1,i}Z_{i+1} + \tau Z_{i+1}\sigma_i F_i,$$

($a_{ii} = 0$ remember), that is

$$a_{i+1,i} = -\frac{\sigma_i F_i \tau}{1 + \tau F_i}.$$

Equating the other random terms, dX_j for $j = 1, \dots, i-1$, gives

$$a_{ij}Z_i = (1 + \tau F_i)Z_{i+1}a_{i+1,j},$$

that is

$$a_{i+1,j} = a_{ij} \quad \text{for } j < i.$$

It follows that

$$a_{i+1,j} = -\frac{\sigma_j F_j \tau}{1 + \tau F_j} \quad \text{for } j < i.$$

Finally, equating the dt terms we get

$$rZ_i = (1 + \tau F_i)rZ_{i+1} + \tau Z_{i+1}\mu_i F_i + \tau\sigma_i F_i Z_{i+1}\sum_{j=1}^{i} a_{i+1,j}\,\rho_{ij}.$$

[2] The drift isn't r, of course, in the real world. But we are in the risk-neutral world for pricing, and in that world all traded instruments have growth r.

From the definition of F_i the terms including r cancel, leaving

$$\mu_i = -\sigma_i \sum_{j=1}^{i} a_{i+1,j}\rho_{ij} = \sigma_i \sum_{j=1}^{i} \frac{\sigma_j F_j \tau \rho_{ij}}{1+\tau F_j}.$$

And we are done.

The stochastic differential equation for F_i can be written as

$$dF_i = \left(\sum_{j=1}^{i} \frac{\sigma_j F_j \tau \rho_{ij}}{1+\tau F_j} \right) \sigma_i F_i \, dt + \sigma_i F_i \, dX_i. \tag{19.9}$$

Equation (19.9) is the discrete BGM version of Equation (19.6) for HJM.[3]

Assuming that we can measure or model the volatilities of the forward rates σ_i and the correlations between them ρ_{ij} then we have found the correct risk-neutral drift. We are on familiar territory, Monte Carlo simulations using these volatilities, correlations and drifts can be used to price interest rate derivatives. In practice, one estimates the volatility functions from the market prices of caplets. Invariably, this is how one calibrates the time-dependent functions, rather than estimating them from historical data, say.

19.17 SIMULATIONS

We can write (19.9) as

$$d(\log(F_i)) = \left(\sigma_i \sum_{j=1}^{i} \frac{\sigma_j F_j \tau \rho_{ij}}{1+\tau F_j} - \tfrac{1}{2}\sigma_i^2 \right) dt + \sigma_i \, dX_i.$$

In simulating this random walk we would typically divide time up into equal intervals, we would assume, in order to integrate this expression from one interval to the next, that the Fs (and the σs and ρs) were all piecewise constant during each time interval. Simulation then becomes relatively straightforward.

What is not so obvious, however, is how to present value the cashflows. We know from the risk-neutral concepts that there are two aspects to calculating the expectation that is the contracts value, and they are

- simulating the risk-neutral random walk;
- present valuing the cashflows.

Well, I've explained the first of these, what about the second?

19.18 PVING THE CASHFLOWS

Before present valuing the cashflows (in anticipation of later averaging, and hence pricing) we must be able to write them in terms of the quantities we have simulated, that is the forward rates. That may be simple or hard. For the simpler instruments they are already defined in terms of these quantities. The more complicated contracts might have cashflows that are, in the sense of our exotic option classification, higher order. In the

[3] Although the notation is slightly different. In the HJM analysis we wrote the random component in terms of uncorrelated dX_i, here we have a different dX_i for each forward rate, but they are all potentially correlated.

HJM framework we present valued these cashflows using the average spot rate r up until each cashflow. In the BGM model we don't have such an r, of course.

In the BGM model we must present value using the discount factors applicable (for each realization) from one accrual period to the next. That is, we present value each cashflow back to the present through all of the dates T_i using the one-period discount factor at each period:[4]

$$\frac{1}{1+\tau F_i(T_i)}.$$

This is the discrete version of the present valuing we do with the average spot rate in the HJM model. Indeed, if you take the limit as $\tau \to 0$ in all of the above equations you will get back to the HJM model, all the sums become integrals, for example.

19.19 SUMMARY

The HJM and BGM approaches to modeling the whole forward rate curve in one go are very powerful. For certain types of contract it is easy to program the Monte Carlo simulations. For example, bond options can be priced in a straightforward manner. On the other hand, the market has its own way of pricing most basic contracts, such as the bond option, as we discussed in Chapter 18. It is the more complex derivatives for which a model is needed. Some of these are suitable for HJM/BGM.

FURTHER READING

- See the original paper by Heath *et al*. (1992) for all the technical details for making their model rigorous.
- For further details of the finite maturity interest rate process model see Sandmann & Sondermann (1994), Brace *et al*. (1997) and Jamshidian (1997).

EXERCISES

1. Derive the equation for the evolution of the forward rate:

$$dF(t;T) = \frac{\partial}{\partial T}\left(\tfrac{1}{2}\sigma^2(t,T) - \mu(t,T)\right)dt - \frac{\partial}{\partial T}\sigma(t,T)\,dX,$$

from

$$dZ(t;T) = \mu(t,T)Z(t;T)dt + \sigma(t,T)Z(t;T)dX,$$

and

$$F(t;T) = -\frac{\partial}{\partial T}\log Z(t;T).$$

[4] Notice that I didn't use the word 'step' instead of 'period' here. 'Step' would refer to the small time step in the Monte Carlo simulation. Period refers to the time between T_{i-1} and T_i.

2. Perform a simulation, using the method of section 19.7, to value an option on a zero-coupon bond. You will need to decide upon a suitable form for $\nu(t, T)$, the forward rate volatility. Does your choice have a standard representation as a model for the spot rate?
3. Using forward rate data, perform a principal component analysis. What are the three main components in the forward price movements and what are their weights?